

Multi-scale intrinsic dimension estimators for text embeddings

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Abstract

Intrinsic dimension (ID) is a useful geometric descriptor of text embeddings, with applications in model analysis, hallucination detection, and AI-generated text detection. Classical ID estimators (e.g., PHDim) produce a single global value, averaging information across all geometric scales. However, text embeddings may exhibit different structure at different scales: local neighborhoods capture lexical repetition and syntax, while global structure reflects semantics and topics.

We propose a family of scale-aware ID estimators based on quantile-trimmed minimum spanning tree (MST) energies. We prove that these estimators preserve the asymptotic scaling law of classical PHDim, yielding consistent estimates at any fixed scale. Using token embeddings from Gemma-4-E4B-it, we evaluate our methods on two detection tasks: the adversarial paraphrase benchmark PadBen and a controlled synthetic benchmark (FLAT) spanning three domains and three generation procedures.

Our multiscale summary med PHDq (median of PHD across scales) outperforms global PHD on PadBen and competes with zero-shot detectors. On FLAT, the interval-trimmed estimator $d_{\text{interval}}(q)$ consistently outperforms or matches global PHD, with the largest gains on long-form texts. Linguistic analysis reveals that different scales capture qualitatively different properties: short edges correlate with lexical richness and style, long edges involve rare tokens and key terms, and intermediate scales form semantically coherent clusters distributed across the text.

Our results demonstrate that scale-dependent ID estimation reveals structure that is averaged out by global estimators, providing a more nuanced geometric view of text representations.

1 Introduction

The geometric structure of text embeddings has become a central object of study in modern NLP, with

applications ranging from model interpretability to AI-generated text detection. One particularly useful geometric characterization is the **intrinsic dimension** (ID) of a set of embeddings – the minimal number of degrees of freedom needed to describe their configuration. Low ID indicates that embeddings lie near a low-dimensional manifold, while high ID suggests more complex, high-dimensional organization.

Traditionally, language modeling and text generation have been evaluated using entropy-based metrics such as perplexity. Perplexity captures uncertainty and predictability at the token level, but it ignores the geometric structure of representations. Intrinsic dimension offers a complementary perspective: it measures how many independent directions are necessary to describe the embedding configuration, regardless of the number of possible surface forms. As shown by [Vladislav et al. \(2026\)](#), a language can be highly ambiguous (many lexical variants) yet still have low intrinsic dimension – meaning that all variants collapse onto a low-dimensional manifold. Thus, perplexity and intrinsic dimension capture fundamentally different aspects of linguistic complexity, and combining them may provide a richer understanding of language models and their outputs.

Classical estimators of intrinsic dimension, such as the persistent homology dimension (PHDim) based on minimum spanning trees (MSTs), produce a single global estimate. However, text embeddings may exhibit different geometric structure at different scales: local neighborhoods might reflect lexical repetition and syntax, while global structure captures semantic topic organization. A single global ID average risks obscuring these scale-specific phenomena.

In this work, we propose a family of **scale-aware intrinsic dimension estimators** based on trimmed MST energies. By selectively including only edges whose lengths lie in a specified quantile

range (lower-trimmed, upper-trimmed, or interval-trimmed), we can probe the geometry at different scales. We prove that these trimmed energies preserve the same asymptotic scaling law as the classical PHDim estimator, and therefore yield consistent estimates of intrinsic dimension for any fixed trimming interval. In our experiments, we primarily use the interval-trimmed version with a sliding window of width 0.5, denoted $d_{\text{interval}}(q)$.

Using token embeddings from a modern language model (Gemma-4-E4B-it), we conduct two sets of experiments.

First, on the PADBEN adversarial paraphrase benchmark, we evaluate a multiscale summary statistic, med PHDq (the median of PHD estimates across scales). Our method outperforms global PHD and competes with zero-shot detectors, especially on challenging tasks such as AI laundering and iterative paraphrase depth.

Second, we introduce the FLAT (Fair Long Artificial Texts) synthetic benchmark, spanning three text domains and three generation procedures. We show that the optimal scale q varies with domain and generation method, and that $d_{\text{interval}}(q)$ consistently outperforms or matches the global PHD estimator. The strongest separation is observed for long-form essays, where the signal increases almost monotonically with q .

Beyond detection, we analyze which linguistic properties are captured at different geometric scales. Short MST edges (low q) correlate with lexical richness, pronoun frequency, and syntactic complexity, while long edges (high q) involve rare tokens, domain-specific terminology, and high-norm embeddings. At intermediate scales, clusters often capture distributed semantic, stylistic, or grammatical patterns across the text – for example, thematic groups like computation, hardware, conventional GPUs or evaluative-dynamic phrases such as computationally inefficient, scaling, dynamics. These findings suggest that geometry at different scales encodes qualitatively different linguistic information.

Our main contributions are:

- A theoretically grounded family of scale-dependent intrinsic dimension estimators based on quantile-trimmed MST energies (Section 3.3).
- A multiscale summary med PHDq that provides strong unsupervised detection on adver-

sarial paraphrase tasks (Section 4.2).

- The FLAT synthetic benchmark, demonstrating that optimal scales vary with text domain and generation procedure (Section 4.3).
- Linguistic analysis linking different MST scales to different textual properties, including the discovery of semantically coherent clusters at intermediate scales (Section 3.4).

The remainder of this paper is organized as follows. Section 2 discusses related work on intrinsic dimension in NLP and topological data analysis. Section 3 presents our multiscale estimators and theoretical guarantees. Section 4 describes the PadBen and FLAT benchmarks and reports experimental results. Section 3.4 analyzes the linguistic properties captured at different scales. Section 5 concludes and outlines future directions.

2 Related Work

Persistent Homology Dimension (PHD) in Neural Networks. PHD combines local and global properties of point clouds (Schweinhart, 2021). Birdal et al. (2021) link it to generalization in neural networks, while Tulchinskii et al. (2023) show that early GPT-generated texts have lower PHD than human-written ones, enabling detection; this gap shrinks for newer models (Kuznetsov et al., 2024). Kushnareva et al. (2024) use local ID changes to detect human/machine boundaries. Recent work applies topological descriptors to LLM representations across layers and training (Gardinazzi et al., 2025; Birdal et al., 2021).

Scale-Dependent PHD Estimation. The MST-based PHDim estimator uses the asymptotic scaling of MST energy with sample size n (Costa and Hero, 2004; Steele, 1988). Standard PHDim aggregates over all scales, obscuring scale-specific phenomena (Schweinhart, 2021). A principled extension fits the growth law over subsampling regimes (Facco et al., 2017; Allegra et al., 2020), but incurs high computational cost and variance (Granata and Carnevale, 2016). Recent advances aim to mitigate this (Chazal et al., 2014; Graf et al., 2026).

Intrinsic Dimension in Neural Networks. ID analysis provides a geometric framework for LLMs. Li et al. (2018) find that good solutions lie in low-dimensional subspaces. Aghajanyan et al. (2021)

show that task-specific variation in pretrained models is low-dimensional, inspiring LoRA (Hu et al., 2022). Recent work shifts to representation space: ID correlates with scaling (Havrilla and Liao, 2024), training saturation (Razzhigaev et al., 2024), entropy and loss (Viswanathan et al., 2025). ID is used for architecture search (He et al., 2023), memorization (Arnold, 2025), and compositional semantics (Lee et al., 2025). Local ID patterns reflect training dynamics and overfitting (Ruppik et al., 2025), and differ between in-context learning and fine-tuning (Janapati and Ji, 2025).

Data-Centric View of Intrinsic Dimension. Narayanan and Mitter (2010) connect dataset ID to sample complexity, empirically confirmed on images (Pope et al., 2021). Tsukagoshi and Sasano (2025) show that task-specific embeddings can be aggressively reduced. Yin et al. (2024) use Local ID (LID) for hallucination detection: high and variable LID indicates fabricated content. Ruppik et al. (2025) show that ID varies locally: common tokens have low LID, rare terms high LID, and higher layers show increased LID and variance. Related work extends geometric methods to internal states for hallucination detection (Zhang et al., 2026; Song et al., 2026).

Thus, ID is a practical tool for interpretability and downstream applications.

3 Multi-scale Persistent Homology Dimension

3.1 PH_0 and minimum spanning trees

Persistent homology characterizes the geometry of a point cloud by analyzing how its connectivity changes across scales. In the case of zero-dimensional persistent homology (PH_0), one starts from isolated points and gradually connects nearby points as the distance threshold increases, tracking how connected components merge.

For a finite point cloud $X_n = \{x_1, \dots, x_n\} \subset \mathbb{R}^d$, a fundamental result states that this merging process is equivalent to constructing the minimum spanning tree (MST). Specifically, if one runs Kruskal’s algorithm and adds edges in increasing order of length, each newly added edge merges two connected components exactly as in PH_0 . Therefore, the sequence of component-merging scales coincides with the sequence of MST edge lengths.

Denote sorted MST edge lengths by

$$l_{(1)} \leq l_{(2)} \leq \dots \leq l_{(n-1)}.$$

The corresponding MST α -energy is

$$S_\alpha(X_n) = \sum_{e \in MST(X_n)} |e|^\alpha.$$

Under standard assumptions on the sampling distribution

$$S_\alpha(X_n) = \Theta\left(n^{1-\alpha/d}\right),$$

which implies

$$d = \frac{\alpha}{1-\beta}, \quad \beta = \lim_{n \rightarrow \infty} \frac{\log S_\alpha(X_n)}{\log n}.$$

This gives the classical persistent homology dimension estimator (PHDim).

3.2 Why standard PHDim mixes scales

Although PHDim produces a single intrinsic-dimension estimate, it implicitly aggregates information from multiple geometric scales.

Indeed, since the MST contains exactly $n - 1$ edges

$$\frac{S_1(X_n)}{n-1} \sim n^{-1/d},$$

meaning that the characteristic edge length decreases with sample size. Larger point clouds therefore reveal progressively finer geometric details.

Consequently, classical PHDim does not correspond to a fixed geometric scale: it averages information over the entire spectrum of MST edges. For objects whose effective geometry changes across scales, this averaging may obscure important structure.

3.3 Scale-dependent trimmed energies

Our goal is to construct a family of scale-aware dimension estimators while preserving the asymptotic guarantees of classical PHDim. We introduce three trimmed versions of MST energy for further analysis.

Lower-trimmed energy removes the shortest edges:

$$S_{\alpha,p}^{lower} = \sum_{i > \lfloor p(n-1) \rfloor} l_{(i)}^\alpha.$$

Upper-trimmed energy removes the longest edges:

$$S_{\alpha,p}^{upper} = \sum_{i \leq \lfloor p(n-1) \rfloor} l_{(i)}^\alpha.$$

Finally, interval-trimmed energy keeps only a quantile range:

$$S_{\alpha,[p_1,p_2]} = \sum_{i=\lfloor p_1(n-1) \rfloor}^{\lfloor p_2(n-1) \rfloor} l_{(i)}^\alpha.$$

Unlike classical PHDim, these constructions allow us to selectively emphasize different parts of the MST spectrum. The critical question is whether such modifications remain theoretically valid.

Theorem 1 (Scale-invariance of trimmed MST energies). *Let X_n be an i.i.d. sample from a compact subset of $\mathbb{R}^d$ with density bounded away from zero and infinity.*

For any fixed trimming parameters

$$S_{\alpha,p}^{\text{lower}}, \quad S_{\alpha,p}^{\text{upper}}, \quad S_{\alpha,[p_1,p_2]}$$

all satisfy

$$S_{\alpha,p} = \Theta\left(n^{1-\alpha/d}\right).$$

Consequently,

$$d_{\text{energy}}(p) = \frac{\alpha}{1 - \lim_{n \rightarrow \infty} \frac{\log S_{\alpha,p}}{\log n}}$$

consistently estimates intrinsic dimension. Thus trimmed energies preserve the asymptotic scaling exponent of classical PHDim and define a valid family of scale-dependent dimension estimators.

Proof sketch. The proof follows from the Penrose–Yukich law of large numbers for stabilizing Euclidean graph functionals. We show that quantile trimming is asymptotically equivalent to threshold truncation with threshold proportional to $n^{-1/d}$. The complete proof is given in Appendix A.

Toy example: scale-dependent geometry. Consider a point cloud consisting of a one-dimensional structure surrounded by weak high-dimensional noise. Short MST edges mostly connect nearby noisy points and therefore reflect local high-dimensional structure, while long edges capture the global low-dimensional skeleton.

Classical PHDim averages both effects into a single estimate. Lower-trimmed energy suppresses local noisy structure and emphasizes the global skeleton, whereas upper-trimmed energy behaves oppositely.

Fig. 1 illustrates this effect on synthetic examples: objects with constant geometry produce nearly flat dimension profiles, whereas multiscale objects exhibit clear transitions between geometric regimes. Although different estimators emphasize different regions of the MST spectrum, Theorem 1 shows that they preserve the same asymptotic scaling law.

3.4 What textual properties are reflected by embedding geometry?

We next ask which linguistic properties are reflected by multiscale geometry in token representations. We denote by $d_{\text{interval}}(q)$ the intrinsic dimension estimate obtained from the interval-trimmed MST energy $S_{\alpha,[q,\min(1,q+0.5)]}$ as defined in Section 3.3.

For each $d_{\text{interval}}(q)$, we compute correlations with a broad set of linguistic features across text-length bins and analyze how these relationships change with q .

Correlations with linguistic features. Fig. 2 shows that different linguistic signals become associated with different parts of the MST spectrum. Lower q (short-to-medium edges) are strongly correlated with lexical and stylistic properties, including lexical richness (MTLD), pronoun frequency, formality, and syntactic complexity. These correlations systematically weaken as q increases (i.e., as we focus on longer edges).

For example, MTLD exhibits a strong positive correlation at small q ($\rho \approx 0.42$), while pronoun frequency reaches $\rho \approx 0.46$. In contrast, features such as long-word ratio, parenthetical usage, and formality index show strong negative correlations. All effects gradually decay toward larger q , indicating that conventional linguistic descriptors align primarily with local and intermediate geometric organization.

Dependence on text length. Fig. 8 in Appendix illustrates how the correlation with MTLD depends on both q and text length. The correlation increases substantially with text length: for short texts the effect is weak, whereas for long texts it reaches $\rho \approx 0.4$. At the same time, increasing q systematically suppresses the correlation: local estimators ($q = 0$) retain the strongest association, while large- q estimators produce substantially weaker correlations. This suggests that longer texts provide sufficient context for local geometric structure to encode richer lexical organization. At larger scales (high q), geometry shifts toward more global and semantically distinctive structures that are less directly captured by conventional linguistic features.

Edge types across quantile bins. To understand what different q capture, we first analyze the edges themselves. We split the MST edges into deciles (bins $[q, q + 0.1]$) and examine the types of connections they represent. Fig. 3 shows that at low

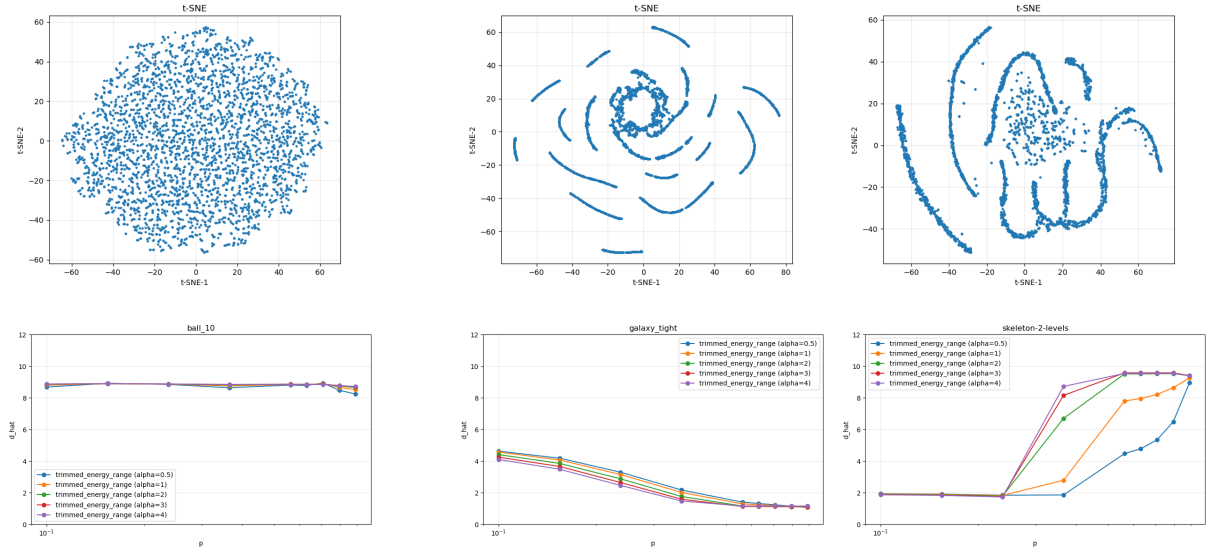


Figure 1: Examples of scale-dependent geometry and corresponding multiscale dimension estimates. Columns show three synthetic datasets: a homogeneous high-dimensional object (left), a low-dimensional structure embedded into noisy high-dimensional geometry (middle), and a sparse multiscale object (right). The first row visualizes data geometry using t-SNE. The second row shows dimension estimates obtained using interval-trimmed MST energies. For homogeneous objects the estimated dimension remains approximately constant across scales. For multiscale objects, different regions of the MST spectrum exhibit distinct scaling behavior, revealing local and global geometric structure that would otherwise be averaged by classical PHDim. The complete visualization, including barcode curves and other trimming strategies, is provided in Appendix B.

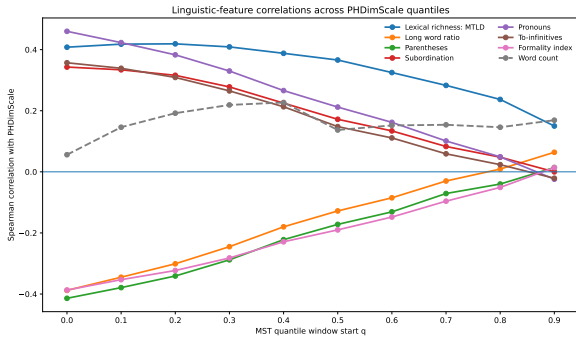


Figure 2: Correlations between PHDimScale and representative linguistic features across MST quantiles. Different linguistic properties dominate at different scales, while text length itself remains only weakly associated with PHDimScale.

q (shortest edges), identical tokens and very short intra-word or intra-phrase connections dominate. As q increases, the proportion of such local edges decreases. After $q \approx 0.5$, their proportion stabilizes, while very short-span edges (within a word or short phrase) slightly increase. Longer edges (high q) increasingly connect heterogeneous, distant parts of the text.

Cluster formation across scales. Next, we follow the persistent homology construction: we add MST edges in increasing order of length and count non-singleton clusters formed by edges shorter than

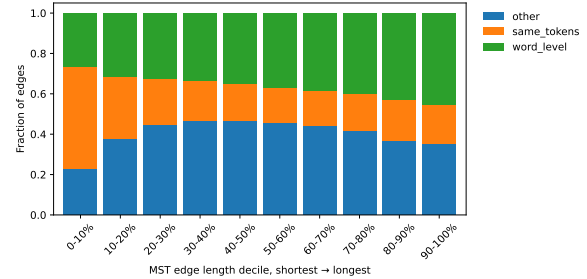


Figure 3: Types of MST edges across edge-length deciles. Shortest edges frequently connect identical tokens or short-span word-level units, whereas longer edges increasingly correspond to heterogeneous long-range connections.

the current quantile q (i.e., edges with length $< q$). Fig. 4 shows the result. The number of clusters first increases (formation of many small local clusters) and then decreases as clusters merge into a giant component. Importantly, the proportion of homogeneous and short-span (within 4 tokens) clusters remains above 2/3 of all clusters across all q , indicating that the unattached clusters correspond to isolated tokens, words, or short phrases.

Semantic, stylistic and grammatical clusters at intermediate scales. At intermediate quantiles, clusters are not limited to repeated tokens or short phrases. Instead, they often capture distributed se-

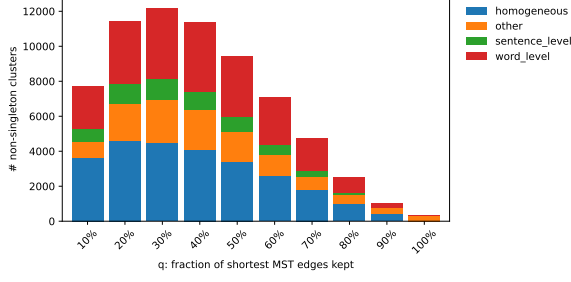


Figure 4: Non-singleton cluster types obtained by keeping the shortest fraction q of MST edges. The number of clusters first increases and then decreases, indicating a transition from local cluster formation to merging between previously formed clusters.

mantic, stylistic, or grammatical phenomena across the text. For example, in one of our texts, a single cluster at intermediate q contained the fragments computation, hardware, conventional GPUs – a thematic cluster around computing hardware. Another cluster included computationally inefficient, have proven highly effective, scaling, dynamics — a mixture of evaluative statements and dynamical concepts. A third cluster grouped the prefixes multi-, single-, low-, capturing morphological patterns. These examples suggest that the geometry at intermediate scales encodes not only lexical repetition but also higher-level semantic and stylistic regularities. We provide additional examples in Appendix C.2.

Token norms. Finally, Fig. 5 shows that tokens for which the shortest incident MST edge falls into a high quantile (i.e., the token is not connected to the rest of the tree via a short edge) tend to have substantially larger embedding norms. This suggests that long MST edges are not merely arbitrary semantic links: they often involve high-norm, geometrically isolated tokens.

4 Artificial Text Detection experiments

4.1 Experimental setup

We evaluate whether multiscale intrinsic dimension estimates capture meaningful geometric differences in text representations. All experiments use token embeddings from the Gemma-4-E4B-it model (google/gemma-4-E4B-it). MSTs are constructed on the set of token embeddings using Euclidean distance, and scale-dependent dimension estimates $d_{\text{PHD-S}}(q)$ are computed from quantile-trimmed MST energies $S_{\alpha, [q, 1.0]}$ as defined in Section 3.3.

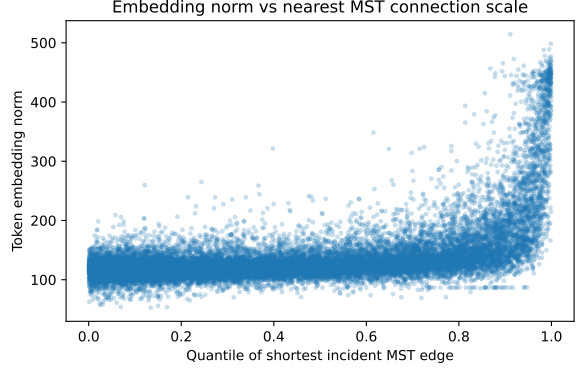


Figure 5: Token embedding norm versus the quantile of the shortest MST edge incident to the token. Tokens that only connect through high-quantile MST edges tend to have larger embedding norms, suggesting that the longest edges often involve geometrically isolated, high-norm tokens.

We consider two complementary detection settings. First, the PADBEN benchmark (Zha et al., 2025) tests robustness against adversarial paraphrase attacks. Second, we construct a synthetic benchmark (FLAT: Fair Long Artificial Texts) spanning multiple text domains and generation procedures; details are given in Appendix D.1.

4.2 Adversarial paraphrases (PadBen benchmark)

PadBen progressively removes stylistic artifacts through paraphrasing and iterative rewriting while preserving semantic content, creating challenging scenarios for AI-text detection (Zha et al., 2025). We use the **exhaustive protocol** (all samples, no resampling) for Tasks 1–5 as defined in the original benchmark.

Since intrinsic dimension estimates do not induce a fixed ordering between human and generated texts, we evaluate separation strength using $|\text{ROC-AUC} - 0.5|$, where 0 indicates random performance and 0.5 indicates perfect separation (in either direction).

Our method. We define med PHD_q as the median of PHD_q (scale-wise persistent homology dimension) estimates across quantile fractions q of the MST edge length distribution. This multiscale summary consistently performs well across tasks.

Comparison with zero-shot and supervised detectors. Table 1 compares med PHD_q against unsupervised zero-shot detectors (Binoculars, Fast-DetectGPT) and, separated by a vertical line, the supervised detector RADAR (Hu et al., 2023). Our method outperforms all unsupervised competitors

Task	med PHDq (ours)	Binoc.	F-DGPT	RADAR*
T1	0.029	0.061	0.073	0.148
T2	0.048	0.079	0.165	0.293
T3	0.062	0.000	0.004	0.133
T4	0.030	0.014	0.012	0.011
T5	0.085	0.072	0.068	0.297

Table 1: $|\text{ROC-AUC} - 0.5|$ on PadBen exhaustive. Our method med PHDq vs unsupervised zero-shot detectors (Binoc. = Binoculars, F-DGPT = Fast-DetectGPT) and supervised RADAR*. Bold = best among unsupervised.

Task	Best ID-based	Binoc.	F-DGPT	RADAR*
T1	energy $p=0.1$ 0.023	0.011	0.006	0.038
T2	energy $p=0.1$ 0.023	0.029	0.010	0.063
T3	energy $p=0.1$ 0.015	0.008	0.011	0.016
T4	med barcode 0.032	0.012	0.009	0.010
T5	med barcode 0.035	0.009	0.006	0.062

Table 2: TPR@1%FPR on PadBen exhaustive. “Best ID-based” is best among global PHD, med PHDq, energy $p=0.1$, med barcode. Binoc. = Binoculars, F-DGPT = Fast-DetectGPT, RADAR* = supervised.

on T3, T4, and T5. On T1 and T2, Fast-DetectGPT is the best.

Low false-positive regime. Table 2 reports True Positive Rate at 1% False Positive Rate (TPR@1%FPR). It illustrates the **best ID-based estimator** (among lobal PHD, med PHDq, energy $p=0.1$ - (e0.1), med barcode) against the same zero-shot and supervised baselines. On T1–T3, energy $p=0.1$ is often the best ID estimator (see Appendix B for details); on T4–T5, med barcode achieves the highest value among ID methods.

Takeaway. Our multiscale estimators based on med PHDq provides strong unsupervised detection across most PadBen tasks, often outperforming zero-shot detectors, especially on challenging tasks like AI laundering (T3) and paraphrase depth (T4).

4.3 The FLAT benchmark: controlled synthetic data

To understand which geometric scales carry discriminative information, we construct the FLAT (Fair Long Artificial Texts) benchmark. FLAT spans three text domains (academic abstracts, news articles, long-form essays) and three generation procedures (direct completion prompting, summary bottleneck, style-preserving rewriting). More details are provided in Appendix D.1.

Multiscale estimator. We use the interval-trimmed MST energy with a fixed window width

Dataset / Model	$d_{\text{interval}}(q)$ (ours)	PHD (global)
<i>Academic abstracts</i>		
deepseek-v4-flash	0.110	0.014
gemini-2.5-flash	0.043	0.013
gpt-5-nano	0.205	0.000
<i>Long-form essays (aeon)</i>		
deepseek-v4-flash	0.353	0.041
gemini-2.5-flash	0.475	0.029
gpt-5-nano	0.301	0.021

Table 3: $|\text{ROC-AUC} - 0.5|$ on the FLAT benchmark for the most challenging generation method (summary_bottleneck_with_style). Our method $d_{\text{interval}}(q)$ (best q per run) is compared against global PHD. Bold indicates the better (higher) value. Our method outperforms global PHD across all configurations. More importantly, while global PHD yields performance close to random guessing, our method consistently provides a substantial predictive signal.

of 0.5:

$$d_{\text{interval}}(q) \text{ computed from } S_{\alpha, [q, \min(1, q+0.5)]}.$$

Here q is the lower quantile of MST edges included in the sum, varying from 0 to 0.9 in steps of 0.1. When $q+0.5 > 1$, the window is truncated to $[q, 1]$. Thus, $q = 0$ emphasizes short-to-medium edges (local geometry), while $q = 0.5$ – 0.9 progressively focuses on long edges (global geometry). Global PHD (using all edges without trimming) is a separate estimator.

Results. Table 3 compares our method (best q per run) against global PHD on the most challenging generation method, summary_bottleneck_with_style. Our method outperforms global PHD in all configurations, often by a large margin, especially on long-form essays (aeon_essays). Full results for all generation methods are provided in Appendix D.3.

Fig. 6 shows $|\text{ROC-AUC} - 0.5|$ as a function of q for different generation models (using the completion_prompting method). The strongest signal typically appears at intermediate q values ($q \approx 0.5$ – 0.8), i.e., when the window focuses on medium-to-long edges. Global PHD (not shown, see Table 3) is consistently weaker.

Dependence on domain and generation procedure. The strongest separation is observed for long-form essays, where the signal grows almost monotonically with q . Academic abstracts and news articles show weaker effects (see Appendix D.2 for details). Results for other

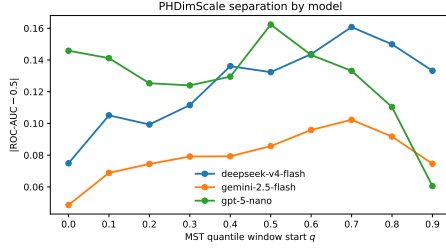


Figure 6: FLAT benchmark: separation ($|\text{ROC-AUC} - 0.5|$) vs q for completion_prompting generation. Our estimator $d_{\text{interval}}(q)$ uses a sliding window of width 0.5. Global PHD values are shown in Table 3.

generation methods (completion_prompting, summary_bottleneck) are provided in Appendix D.3.

Takeaway. Text representations contain discriminative information concentrated at specific geometric scales rather than uniformly across all scales. Our interval-trimmed estimator $d_{\text{interval}}(q)$ reveals structure that is partially averaged out by global intrinsic dimension estimates.

5 Conclusion

Geometric (differential-geometric) methods are difficult to apply to real-world data, because most of them assume a smooth manifold structure that is hard to expect from natural data – especially text. Indeed, texts exhibit highly non-uniform token distributions, a long tail of rare words, and inherently multiscale organization: the structure at the sentence level is qualitatively different from that at the word level.

Methods based on intrinsic dimension have shown promising results, but they are often unstable. One reason is that very different phenomena can affect dimension in similar ways. For example, a shift to a more formal style and end-of-generation looping may both influence the estimated dimension. The PHDim estimator is attractive because it relies on both local and global information from the point cloud, but this same property makes it difficult to use: signals from different scales are mixed together.

In this work, we made an attempt to separate signals from different scales. We provided a mathematically rigorous framework that is valid for data of any nature, proving that quantile-based trimming (lower, upper, or interval) preserves the asymptotic scaling law of the MST energy, and therefore yields consistent intrinsic dimension estimators at any

fixed scale. Unlike heuristic or post-hoc multiscale analyses, our estimators are theoretically grounded and applicable to any point cloud data.

Focusing on text data, we then showed what different scales correspond to linguistically: *low scales* (short MST edges) capture frequent words and short phrases, *intermediate scales* reflect semantic and discourse structure, and *high scales* (long edges) involve key terms, document-specific terminology, and repeated non-standard patterns.

We demonstrated that this multiscale approach distinguishes human-written from generated texts better than classical PHDim, and achieves competitive results on challenging adversarial paraphrase tasks (PadBen) as well as on our synthetic FLAT benchmark.

We hope that our work lays a foundation for a deeper understanding of the geometric structure of neural text representations, and opens the door to a wider range of practical applications related to LLM safety and interpretability.

6 Limitations

While our study provides several insights into scale-dependent intrinsic dimension estimation for text embeddings, it has a number of limitations that should be acknowledged.

Computational cost. Computing MSTs for long texts (e.g., essays of up to 3000 tokens) is more expensive than single global PHDim, especially when evaluating multiple quantile windows q . However, the overhead is manageable for offline analysis.

Embedding dependence. All experiments use only one embedding model (Gemma-4-E4B-it). It remains an open question how our findings generalize to other encoders, especially sentence-level or multilingual embeddings.

Language and domain coverage. We focus exclusively on English texts and three domains (academic abstracts, news articles, long-form essays). The behavior of scale-dependent ID estimators in other languages or genres is not explored.

Stability of trimmed estimators. We did not systematically study the stability of our estimators with respect to the amount of trimming. In particular, it is not yet known what is the minimum number of edges that must remain after trimming (i.e., the smallest allowed quantile window) for the asymptotic scaling law to hold in finite-sample

settings. This is an important direction for future theoretical and empirical analysis.

Detection performance. Our detection experiments on PadBen and FLAT do not show a large margin over existing competitors or extremely high absolute separation scores. We believe that one reason is that our method is most suitable for specific types of data and, more importantly, requires careful hyperparameter tuning (e.g., choosing the right quantile window or the optimal q for a given task). Developing automatic or task-adaptive selection of scales is left for future work.

Diagnostic, not production. Our detection experiments on PadBen are designed as a diagnostic (the portfolio of ID estimators), not as a production-ready detector. Supervised methods such as RADAR achieve stronger absolute performance, but they require task-specific training.

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A Proofs

A.1 Penrose–Yukich LLN for stabilizing graph functionals

To derive scale-calibrated truncations, we use a general law of large numbers (LLN) for Euclidean functionals on random graphs (Penrose–Yukich LLN), which allows us to derive more topologically sound scale-dependent MST-based dimensionality estimators.

Let $G(\mathcal{X})$ be a random geometric graph constructed on a locally finite set $\mathcal{X} \subset \mathbb{R}^d$. Assume that: (i) G is translation-invariant and scale-invariant; and (ii) G stabilizes on a homogeneous Poisson process, meaning that the local edge configuration incident to a typical vertex becomes insensitive (almost surely) to modifications of points outside some random but finite radius.

Given a measurable edge functional $\phi : [0, \infty) \rightarrow \mathbb{R}$, define

$$L_\phi^G(\mathcal{X}) := \sum_{e \in G(\mathcal{X})} \phi(|e|).$$

For an i.i.d. sample $X_n = \{X_1, \dots, X_n\}$ with density f in $\mathbb{R}^d$, the Penrose–Yukich LLN states (under suitable moment conditions) that

$$\begin{aligned} \lim_{n \rightarrow \infty} \frac{1}{n} L_\phi^G(n^{1/d} X_n) &= \\ &= \frac{1}{2} \int_{\mathbb{R}^d} \mathbb{E} \left[\sum_{e \in E(0; G(P_{1,0}))} \phi\left(\frac{|e|}{f(x)^{1/d}}\right) \right] f(x) dx, \end{aligned} \quad (1)$$

where $P_{1,0}$ is the Palm version of a unit-intensity Poisson process in $\mathbb{R}^d$ (a unit Poisson process conditioned to have a point at the origin), and $E(0; G(P_{1,0}))$ denotes the set of edges of $G(P_{1,0})$ incident to the origin. The factor $\frac{1}{2}$ appears because summing incident-edge contributions over vertices counts each undirected edge twice.

In our setting, $G = \text{MST}$; the MST is a canonical example of a stabilizing Euclidean graph, so (1) applies to a broad class of functions ϕ . The exact form of the right-hand side is not important for our purposes.

A.2 Three choices of ϕ and two scale-dependent dimension estimators

Throughout, let $X_n = \{X_1, \dots, X_n\} \subset \mathbb{R}^d$ be an i.i.d. sample, and let $\text{MST}(X_n)$ be its Euclidean minimum spanning tree. For an edge e we write $|e|$ for its Euclidean length.

Functional 1: polynomial weight. Let $\phi_\alpha(u) := \frac{1}{2}|u|^\alpha$, $\alpha > 0$. Substituting ϕ_α into (1) gives

$$\begin{aligned} \frac{1}{n} L_{\phi_\alpha}^{\text{MST}}(n^{1/d} X_n) &= \\ &= n^{\alpha/d-1} \sum_{e \in \text{MST}(X_n)} |e|^\alpha \longrightarrow C_\alpha(f, d), \end{aligned}$$

and therefore

$$\sum_{e \in \text{MST}(X_n)} |e|^\alpha = \Theta(n^{1-\alpha/d}), \quad (2)$$

which is exactly the MST energy scaling.

Functional 2: counting edges above a threshold.

Let $\phi_t^{\text{cnt}}(u) := \frac{1}{2} \mathbf{1}\{|u| > t\}$, $t > 0$. This counts edges longer than threshold t . Then (1) yields

$$\begin{aligned} \frac{1}{n} L_{\phi_t^{\text{cnt}}}^{\text{MST}}(n^{1/d} X_n) &= \\ &= \frac{1}{n} \# \{e \in \text{MST}(X_n) : |e| > t n^{-1/d}\} \\ &\longrightarrow C_{\text{cnt}}(t; f, d). \end{aligned}$$

Define the tail edge-length fraction

$$b_n(x) := \frac{1}{n} \# \{e \in \text{MST}(X_n) : |e| > x\}.$$

The above convergence can be written compactly as $b_n(t n^{-1/d}) \rightarrow C_{\text{cnt}}(t; f, d)$. Equivalently, consider the inverse (quantile) function

$$b_n^{-1}(p) := \inf\{x \geq 0 : b_n(x) \leq p\}, \quad p \in (0, 1),$$

which is the edge length threshold such that a fraction p of MST edges are longer than $b_n^{-1}(p)$. The stabilization of $b_n(t n^{-1/d})$ implies that for each fixed $p \in (0, 1)$,

$$\begin{aligned} b_n^{-1}(p) &= \Theta(n^{-1/d}), \\ \text{hence} \quad \frac{\log b_n^{-1}(p)}{\log n} &\rightarrow -\frac{1}{d}. \end{aligned} \quad (3)$$

This provides a scale-dependent dimension estimator based on fixed-rank (fixed-quantile) MST edge lengths.

A further useful implication is that *length-threshold truncation* at $t n^{-1/d}$ is asymptotically equivalent to *quantile truncation*: removing edges shorter than $t n^{-1/d}$ removes approximately a fixed fraction $p = C_{\text{cnt}}(t; f, d)$ of the smallest edges.

Functional 3: truncated polynomial weight.

Let $\phi_{\alpha,t}(u) := \frac{1}{2}|u|^\alpha \mathbf{1}\{|u| > t\}$, $\alpha > 0$, $t > 0$. Substituting into (1) gives

$$\begin{aligned} \frac{1}{n} L_{\phi_{\alpha,t}}^{\text{MST}}(n^{1/d} X_n) &= \\ n^{\alpha/d-1} \sum_{e \in \text{MST}(X_n)} |e|^\alpha \mathbf{1}\{|e| > tn^{-1/d}\} & \\ \longrightarrow C_{\alpha,t}(f, d). \end{aligned}$$

Therefore the *length-truncated MST α -energy*

$$S_{\alpha,t}(X_n) := \sum_{e \in \text{MST}(X_n)} |e|^\alpha \mathbf{1}\{|e| > tn^{-1/d}\}$$

satisfies

$$S_{\alpha,t}(X_n) = \Theta(n^{1-\alpha/d}), \quad (4)$$

so

$$\frac{\log S_{\alpha,t}(X_n)}{\log n} \rightarrow 1 - \frac{\alpha}{d}. \quad (5)$$

Using the equivalence between length-threshold and quantile truncation, one may also define the *quantile-trimmed α -energy*

$$S_{\alpha,p}(X_n) := \sum_{i=\lfloor p(n-1) \rfloor + 1}^{n-1} \ell_{(i)}^\alpha,$$

where $\ell_{(1)} \leq \dots \leq \ell_{(n-1)}$ are MST edge lengths in ascending order (i.e., we drop the shortest p -fraction of edges and sum the remainder). For fixed $p \in (0, 1)$, $S_{\alpha,p}(X_n)$ obeys the same scaling exponent $1 - \alpha/d$.

Two scale-dependent estimators and their barcode interpretation. Equations (3) and (4) yield two families of scale-dependent dimension estimators:

$$\hat{d}_{\text{quant}}(p) := - \left(\lim_{n \rightarrow \infty} \frac{\log b_n^{-1}(p)}{\log n} \right)^{-1}, \quad (6)$$

$$\hat{d}_{\text{energy}}(\alpha, t) := \frac{\alpha}{1 - \lim_{n \rightarrow \infty} \frac{\log S_{\alpha,t}(X_n)}{\log n}}. \quad (7)$$

Both are naturally expressed in terms of the normalized PH_0 barcode. Sorting MST edges in *descending* order $d_{(1)}(n) \geq \dots \geq d_{(n-1)}(n)$, the barcode envelope $b_n^{\text{env}}(u) := d_{(\lceil u(n-1) \rceil)}(n)$, $u \in (0, 1)$, is exactly the fixed-rank edge length, corresponding to $b_n^{-1}(p)$ (up to the $u \leftrightarrow p$ convention). Moreover, the quantile-trimmed energy $S_{\alpha,p}(X_n)$ equals the sum of α -powers of all bars before rank fraction p . Thus, if we overlay barcodes for multiple n and

examine either (i) the fixed-rank lengths $b_n^{\text{env}}(u)$ or (ii) the cumulative α -energy of the longest bars, we obtain a direct visual and quantitative probe of scale dependence: if the intrinsic dimension is constant across scales, the corresponding log–log “level lines” across n are asymptotically parallel; abrupt changes in effective dimension manifest as a clear change of slope at the associated rank fractions.

Empirical illustrations (overview). In the experiments below we: (i) construct several synthetic multiscale point clouds with known changes of effective dimension across scales; (ii) for each dataset, plot log–log curves of the full MST energy $S_\alpha(X_n)$, the length-truncated energies $S_{\alpha,t}(X_n)$ for multiple t , the quantile-trimmed energies $S_{\alpha,p}(X_n)$ for multiple p , and the fixed-rank edge lengths $b_n^{-1}(p)$; (iii) overlay the normalized PH_0 barcode envelopes $b_n^{\text{env}}(u)$ across n to visualize the “barcode relief map” described above.

Fig. 7 demonstrates that all our estimators are stable on a simple object (a 10-dimensional ball) and are able to capture the multi-scale nature of more complex point clouds. Importantly, in some cases certain estimators fail to detect scale-wise changes in dimensionality (e.g., trimming the shortest edges does not help when there is tight high-dimensional noise around 1-dimensional structures on a disk), whereas their combination is effective: in the above example, both the length-based predictor and the high-edge trimming reveal this dimensionality increase, although they underestimate its absolute value.

B Synthetic examples

B.1 Connection to barcode statistics

The collection of birth–death intervals is called a *barcode*. In dimension zero, all births occur at zero, therefore the barcode is completely determined by component death times.

The sorted MST edge lengths define barcode curves describing how persistent scales are distributed across the tree.

$$b_n(u) = l_{(\lceil u(n-1) \rceil)}, \quad u \in (0, 1).$$

Lower trimming accumulates contributions from long bars and therefore emphasizes coarse global structure. Upper trimming instead focuses on local geometry encoded by short bars, while interval trimming captures intermediate scales.

Consequently, the resulting dimension profiles can be interpreted directly as measurements over different regions of the barcode spectrum.

Fig. 7 illustrates this behavior on synthetic examples.

C Embeddings geometry

C.1 Linguistic feature correlation vs text length

See Fig. 8 for details.

C.2 Examples of homogeneous MST clusters

Table 4 shows representative homogeneous clusters sampled at different MST quantiles. Low-quantile clusters are dominated by punctuation and frequent function words. Intermediate quantiles contain repeated content words and short lexical units. High quantiles increasingly contain rare subword fragments, names, symbols, and domain-specific pieces.

C.3 Qualitative examples across MST scales

To better understand what different MST scales capture, we visualize connected token clusters for the same text at two different quantile ranges.

Fig. 9 compares clusters obtained from only the shortest MST edges ($q < 0.3$) and from only the longest edges ($q > 0.7$).

At small scales, clusters often remain locally coherent and reveal meaningful thematic structure. Even when BPE tokenization splits words into fragments, neighboring clusters still contain recognizable semantic concepts such as *window*, *global*, *noise*, *clients*, or *federated*. Thus local geometry appears to organize tokens into compact semantic regions.

At larger scales, the behavior changes substantially. The most prominent clusters increasingly involve fragments of multi-token expressions and domain-specific terminology, such as pieces of *federated*, *heterogeneous*, or benchmark names like *CIFAR*. Rather than representing local lexical similarity, these edges frequently connect sparse and informative regions of the text.

This observation suggests a transition between two regimes. Lower MST scales primarily capture local semantic organization and lexical repetition, whereas higher scales increasingly connect previously formed semantic regions through rare or highly informative textual elements.

D AI-generated text detection

D.1 FLAT dataset

To construct a robust benchmark for AI-generated text detection, we introduce **FLAT**, a multi-domain, multi-model dataset. We curated human-authored texts by randomly sampling 1,000 instances each from three distinct domains: academic abstracts¹, long-form essays from Aeon Essays², and news articles from CNN/DailyMail³. For every human text, we utilized gpt-4.1-mini to extract two levels of metadata: a concise summary capturing the core ideas, and a brief stylistic description outlining formatting and textual markers. AI counterparts were then generated across three language models (deepseek-v4-flash, gemini-2.5-flash, and gpt-5-nano) using three distinct prompting strategies per model-domain pair. First, *completion prompting* tasked the models with continuing a text based on its original title and the first 50 words. Second, *summary bottleneck* prompting required generating a text solely from the extracted core summary. Third, *summary bottleneck with style* augmented the summary with the stylistic description to guide the generation. The resulting **FLAT** dataset provides a diverse and strictly controlled environment specifically designed to evaluate detector robustness against varied prompting constraints.

The benchmark is designed with a strict focus on ensuring a fair comparison between AI-generated and human-authored texts. Similarity in topic and style distributions is inherently enforced by the generation process, since each AI-generated text is derived from a corresponding human source text. In addition, we make efforts to ensure a fair length distribution: generation prompts explicitly instruct models to produce texts with exactly the same number of words as their human counterparts. Figure 10 demonstrates that models were generally able to follow these instructions, with the exception of essays, where they exhibited a strong tendency toward a preferred output length.

D.2 Detailed analysis of FLAT performance by domain

Fig. 11 shows separation performance for different text domains. Long-form essays exhibit by

¹<https://www.kaggle.com/datasets/beta3logic/3m-academic-papers-titles-and-abstracts>

²<https://www.kaggle.com/datasets/mannacharya/aeon-essays-dataset>

³https://huggingface.co/datasets/abisee/cnn_dailymail

Scale	Typical cluster content	Examples
Low q	punctuation / function words / common discourse markers	. . ., the the, to to, he he he, said said, (((, cent cent, You you
Middle q	repeated lexical items and domain-level semantic phrases	code code code, shot shot shot, genomic genomic genetic, Language Language, proportion number, associated contributing, picture left right, initial The
High q	rare fragments / names / domain-specific terminol- ogy	narr narr, curv curv, Se Se, X X, Duch Duch, Mani Mani, propio proprio, EW EW, No significant differences

Table 4: Representative homogeneous clusters across MST scales. Low quantiles are dominated by frequent syntactic tokens and punctuation, middle quantiles contain repeated lexical or semantic patterns, while high quantiles capture rare domain-specific fragments, named entities, and specialized terminology.

far the strongest multiscale effect: separation increases substantially with increasing quantile level and reaches its maximum at large scales. Academic abstracts and news articles show weaker effects and flatter profiles.

D.3 Detailed analysis of FLAT performance by model and method

Fig. 6 breaks down results by generation model. Despite differences in overall performance, all models display similar qualitative behavior: informative signals emerge primarily at intermediate and large quantiles rather than at the global scale. The maxima occur consistently around $q \approx 0.5 - 0.8$, indicating that the relevant geometric structure is relatively stable across model families.

Fig. 12 shows the dependence on generation strategy. Direct completion prompting produces the strongest separation, while summary-based methods and style-preserving transformations are more difficult to distinguish. Nevertheless, all methods exhibit a similar scale dependence, with performance degrading near both extreme ends of the quantile range.

Taken together, these results suggest that the geometric scale carrying useful information is substantially more stable than individual datasets or generation procedures. The main variability arises in signal strength rather than in the location of the informative scales.

E Technical details

E.1 Computation resources

For our experiments we used servers with the following computational resources:

- 2 x NVIDIA GeForce RTX 4090 GPU + Intel(R) Core(TM) i7-14700K CPU

- 4 x NVIDIA A100 80 Gb + Intel(R) Xeon(R) Gold 6348 CPU @ 2.60GHz

E.2 Artifact licenses

- PadBen dataset - The MIT License

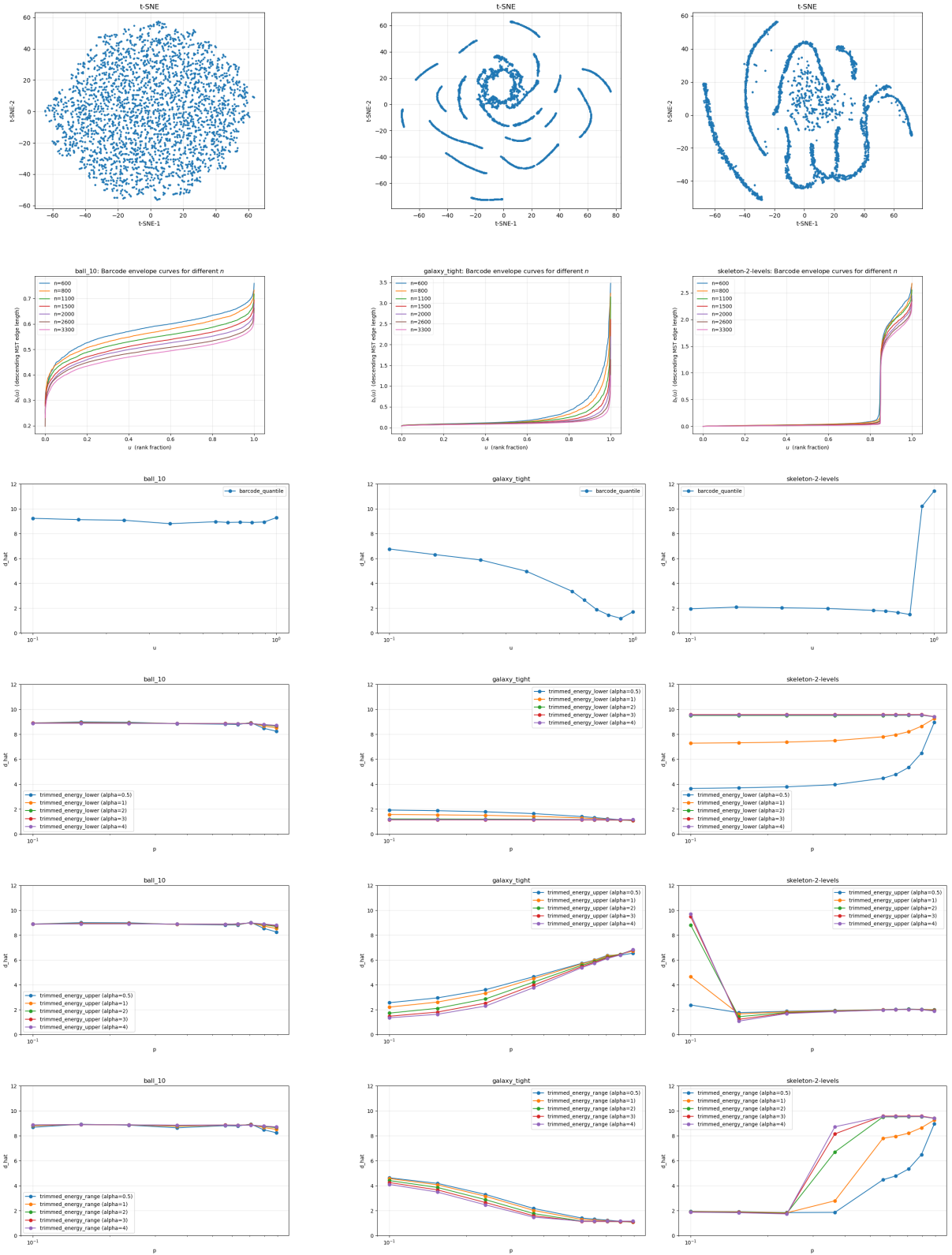


Figure 7: Examples of scale-dependent geometry and corresponding multiscale dimension estimates. Columns show three synthetic datasets: a homogeneous 10-dimensional ball (left), a low-dimensional structure with local high-dimensional perturbations (middle), and a sparse multiscale object (right). Rows visualize: data geometry, barcode functions, fixed-rank estimates, lower-trimmed energies, upper-trimmed energies, and interval-trimmed energies. Classical PHDim averages information across scales, whereas trimmed estimators reveal different geometric structures while preserving the same asymptotic scaling law.

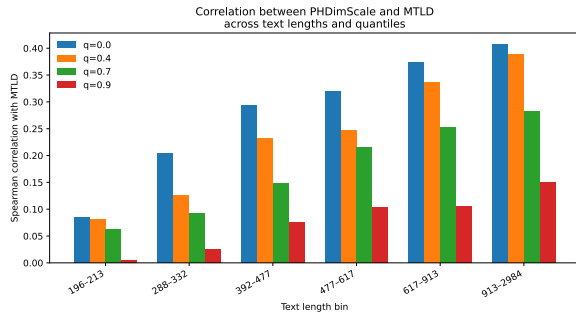


Figure 8: Dependence of the correlation between PHDimScale and lexical richness (MTLD) on text length for different MST quantiles. The relationship strengthens substantially for longer texts, while larger MST quantiles systematically reduce the effect.

Window-based Model Averaging Improves Generalization in Heterogeneous Federated Learning

text_id=4aee4096b3a0 | dataset=academic_abstracts | source=generated | q=0.3

<bos> Federated Learning (FL) aims to learn a global model from distributed users while protecting their privacy. However, when data are distributed heterogeneously the learning process becomes noisy, unstable, and biased towards the last seen clients' data, slowing down convergence. To address these issues and improve the robustness and generalization capabilities of federated models, we propose a window-based model averaging strategy. Instead of aggregating only the most recent global model, our method maintains a sliding window of past model states and computes a weighted average over them. This temporal smoothing mitigates the bias introduced by the last client's local update and reduces variance from client sampling. We theoretically analyze the convergence rate under non-i.i.d. data and show that window averaging reduces the impact of gradient noise. Extensive experiments on benchmark datasets (CIFAR-10, FEMNIST, and Shakespeare) with varying degrees of heterogeneity demonstrate that our approach consistently outperforms standard Fed Avg and state-of-the-art robust aggregation methods. In particular, window-based averaging improves test accuracy by up to 4% and accelerates convergence by reducing communication rounds. Moreover, the method is computationally lightweight and requires no additional communication overhead. Our findings suggest that simple temporal ensembling of global models is an effective and practical technique for enhancing generalization in heterogeneous federated learning settings.

(a) Shortest MST edges only ($q < 0.3$)

Representative clusters:

- particular, Moreover, suggest that
- slowing down improve accelerates enhancing
- noisy unstable biased issues robustness
- averaging approach consistently method
- update variance impact noise
- analyze show demonstrate

Window-based Model Averaging Improves Generalization in Heterogeneous Federated Learning

text_id=4aee4096b3a0 | dataset=academic_abstracts | source=generated | q=0.7

<bos> Federated Learning (FL) aims to learn a global model from distributed users while protecting their privacy. However, when data are distributed heterogeneously the learning process becomes noisy, unstable, and biased towards the last seen clients' data, slowing down convergence. To address these issues and improve the robustness and generalization capabilities of federated models, we propose a window-based model averaging strategy. Instead of aggregating only the most recent global model, our method maintains a sliding window of past model states and computes a weighted average over them. This temporal smoothing mitigates the bias introduced by the last client's local update and reduces variance from client sampling. We theoretically analyze the convergence rate under non-i.i.d. data and show that window averaging reduces the impact of gradient noise. Extensive experiments on benchmark datasets (CIFAR-10, FEMNIST, and Shakespeare) with varying degrees of heterogeneity demonstrate that our approach consistently outperforms standard Fed Avg and state-of-the-art robust aggregation methods. In particular, window-based averaging improves test accuracy by up to 4% and accelerates convergence by reducing communication rounds. Moreover, the method is computationally lightweight and requires no additional communication overhead. Our findings suggest that simple temporal ensembling of global models is an effective and practical technique for enhancing generalization in heterogeneous federated learning settings.

(b) Longest MST edges only ($q > 0.7$)

Representative clusters:

- federated learning
- are distributedly biased
- CIFAR-10
- FL datasets (FEMNIST, Shakespeare particular)
- learn a global
- from distributed iid

Figure 9: Qualitative comparison of MST scales for the same text. Shortest MST edges form locally coherent semantic clusters composed of common domain vocabulary. Longest MST edges increasingly involve fragments of multi-token concepts and highly informative terms (e.g., federated learning, distributionally biased, CIFAR-10, IID), acting as bridges between larger semantic regions.

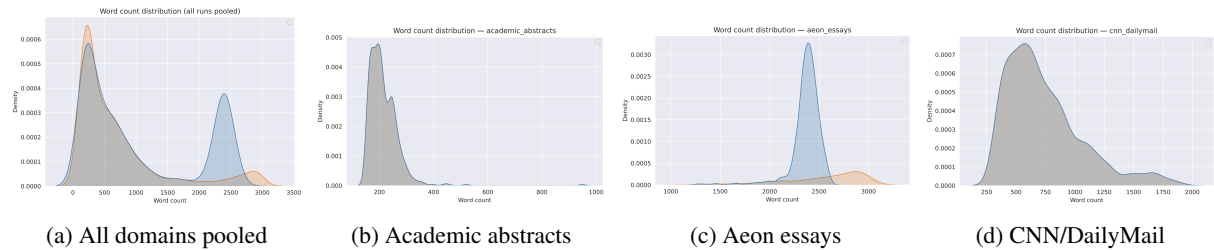


Figure 10: Word count distributions for human and AI-generated texts across domains. The generation process explicitly constrained models to match the length of the corresponding human text. The distributions closely align for academic abstracts and news articles, indicating successful control of text length. In contrast, for essays models exhibit a pronounced tendency toward preferred output lengths, leading to a noticeable shift from the human distribution.

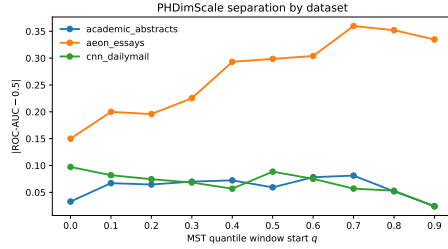


Figure 11: PHDimScale separation across text domains. Long-form essays show the strongest scale-dependent separation, while academic abstracts and news articles exhibit weaker signals.

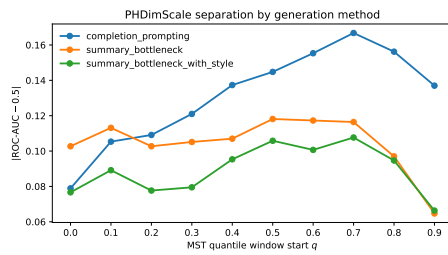


Figure 12: PHDimScale separation across generation methods. All methods show clear dependence on the selected MST quantile range, with informative signal concentrated away from the global estimator.